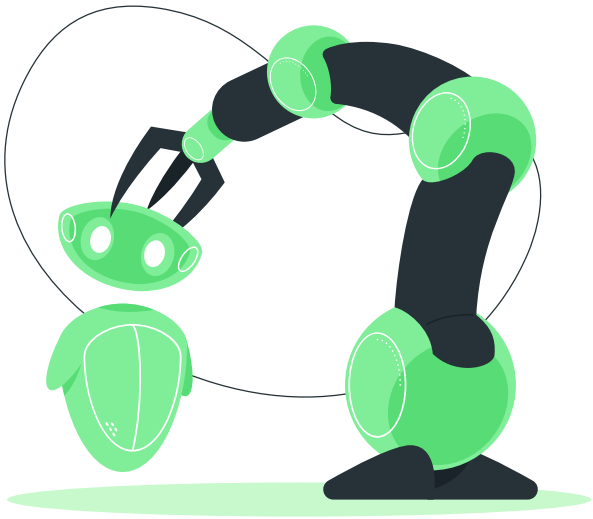
Robotics
CS 3015

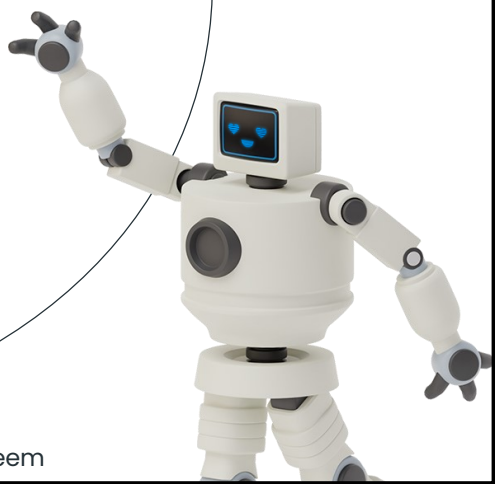
Umarah Qaseem



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- Measuring motion
- Inertial Navigation System
- Inertial Measurement Unit
- Inertial Sensors
- Accelerometer
- Magnetometers
- Gyroscope
- Gimbal Mechanisms
- Gimbal Lock
- Nasa's Apollo IMU



2

Measuring Motion – Sensors

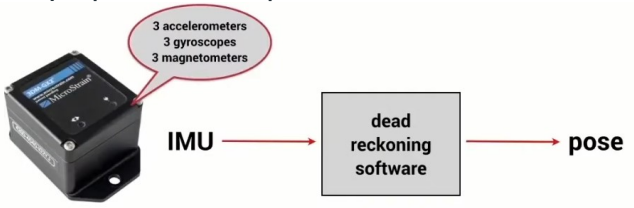


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Inertial Navigation System INS

- Key components are
  - Inertial Measurement Unit (IMU)
  - Dead reckoning software
- It continually updates the pose of the vehicle.



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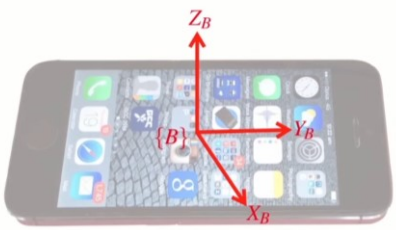
Sensors

- **Gyroscopes**: sensor that measures the rate of change of orientation.
- **Accelerometers:** sensors that measure acceleration.
- **Magnetometers**: sensors measure magnetic field to determine the heading.
- 6-DOF IMU: tri-axial gyroscopes + accelerometers
- 9-DOF IMU: tri-axial gyroscopes + accelerometers + magnetometers

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Three-axis accelerometer

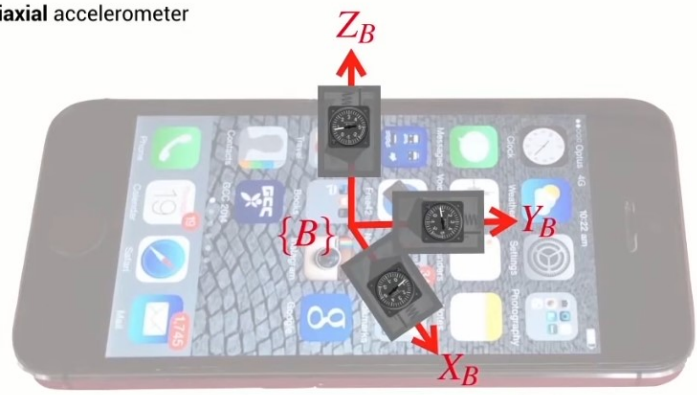
- Search the manual of your phone to find out how it is positioned.
- Determines the roll & pitch angles.



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Three-axis accelerometer

- One accelerometer aligned with each axis of the device
  - a **triaxial** accelerometer



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Three-axis accelerometer

- When sensor is not moving the value is this.
- Why?

$$\boldsymbol{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}$$

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Applications of accelerometers

- airbag trigger
- camera and phone/tablet orientation
- stabilization of aircraft



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Magnetic Compass



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Magnetic Poles of Earth

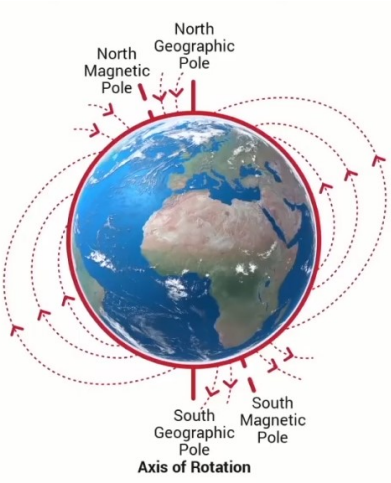
- Constantly moving



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Magnetic Compass

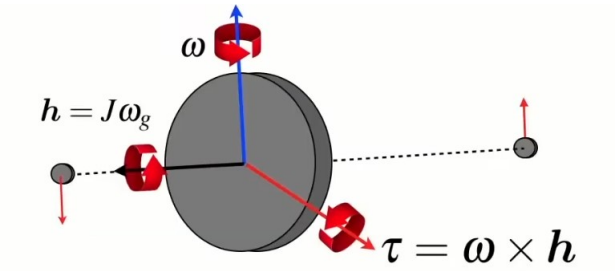
- Compass north points towards the north magnetic pole
- North geographic and magnetic pole are different.
- Yaw angle is calculated.



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Gyroscope

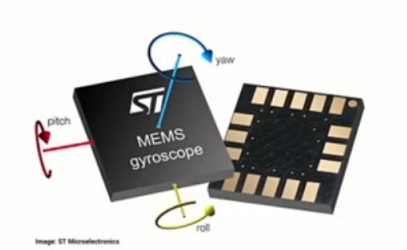
- Rotating the gyroscope causes an output torque proportional to the rotational rate which can be measured.



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Modern day Gyroscopes

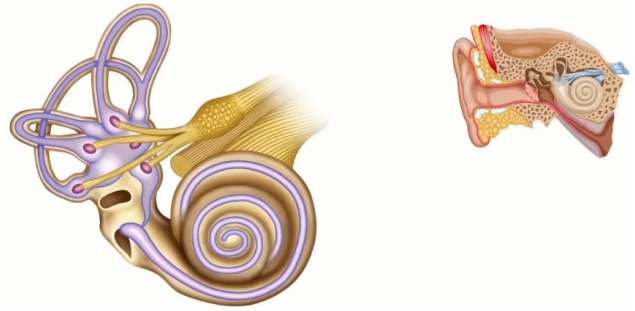
- Uses vibrating rather than rotating elements.



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Biological Equivalents

- 2 accelerometers
- 3 Gyroscopes



Biological Equivalents

- Motion Sickness?
- Semi circular canals have liquid them. The rotation motion of head causes the fluid in them to move, it is detected by tiny hairs cells in the canals. This leads to angular velocity signal to brain. The signal is combined with motion signal information from your eyes. If both these signals are not consistent then it is called motion sickness or sea sickness where our eyes gives information that body is moving one way while the inner ears tell us that our body is moving the other up and our body respond to it illogically by wanting to throw up.

Application: Inertial Navigation System

- Inertial Navigation System(INS) measures accelerations, magnetic field and angular velocities and integrates them over time to estimate
  - Velocity
  - Orientation
  - Position
- No external input making it feasible for applications such as submarine, spacecraft and missile guidance systems etc

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Application: Inertial Navigation System

- INS measure its pose wrt inertial reference frame {0} fixed to some point on earth's surface (world coordinate frame).
- The frame has its plane x-y tangent to earth's surface, whereas z point upwards/downwards.
- **Conventions:**

x-, y- and z-axes	x-, y- and z-axes
north-east-down (NED)	east-north-up (ENU)

The coordinate frame $\{B\}$ is attached to the moving vehicle or robot and is known as the body- or body-fixed frame.

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Inertial Sensors

- **Accelerometers** (gravity)
- **Gyroscopes**: (angular velocity)
- **Magnetometers**: (magnetic field)

- Combine information from accelerometers and magnetometers to get the orientation of body wrt. magnetic north.

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Gimbal Mechanism

- Have you ever watched a drone gliding through the air, capturing video, and wondered how it got such steady footage?
- The answer is a gimbal.
- But what is a gimbal?
- It's essentially a platform that can pivot.
- What does that mean?
- Well, it means that instead of being fixed to an unmoving base, an object on a gimbal can rotate along at least one axis. In the world of robotics/aeronautics, these axes are roll, pitch and yaw.

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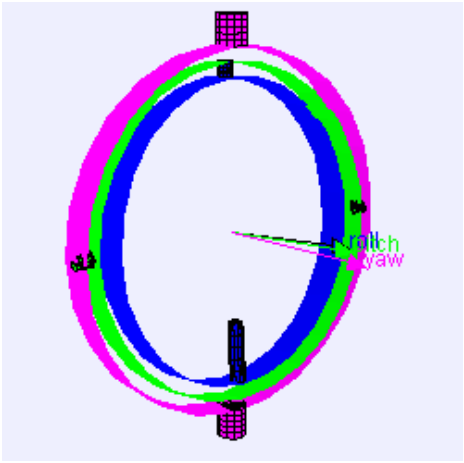
Gimbal Mechanism



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Gimbal Mechanism

- Roll, pitch, yaw



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What is a Gimbal

- A gimbal is a pivoted support that permits rotation of an object about an axis.
- A set of three gimbals, one mounted on the other with orthogonal pivot axes, may be used to allow an object mounted on the innermost gimbal to remain independent of the rotation of its support (e.g. vertical in the first animation in previous slide).

- (source: wiki)

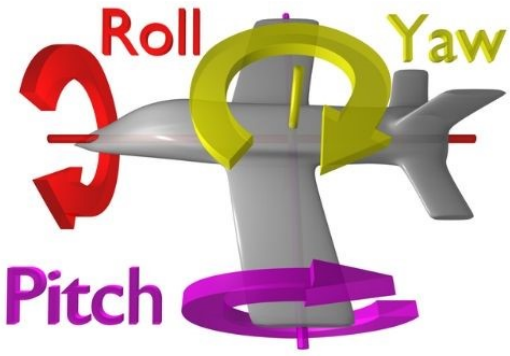
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Use

- Take example of a **ship**:
- The **gyroscopes**, **shipboard compasses**, **stoves**, and even **drink holders** typically use gimbals to keep them upright with respect to the horizon despite the ship's movement (pitching and rolling.)

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Roll, Pitch, Yaw



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Applications

- Inertial navigation
- Rocket engines
- Photography and imaging
- Film and video
- Marine chronometers

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Gimbal Lock

- Gimbal lock is the loss of one degree of freedom in a three-dimensional, three-gimbal mechanism.
- It occurs when the axes of two of the three gimbals are driven into a parallel configuration, "locking" the system into rotation in a degenerate two-dimensional space.

- In easy words, when the rotational axis of one term in the sequence becomes parallel to the rotation axis of the other. **(Singularity)**

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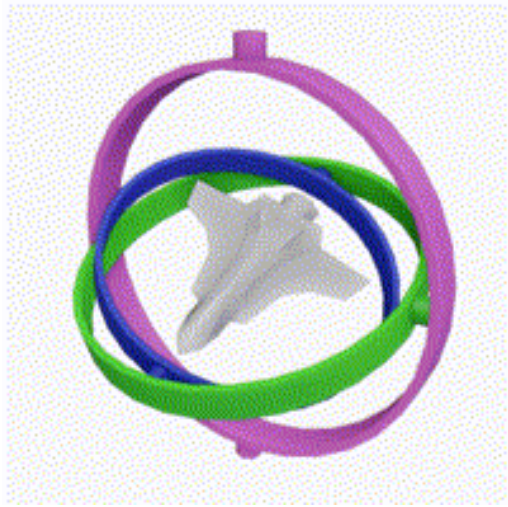
Gimbal Lock/Singularity

- In easy words, when the rotational axis of one term in the sequence becomes parallel to the rotation axis of the other. **(Singularity)**

- The word lock is misleading: no gimbal is restrained. All three gimbals can still rotate freely about their respective axes of suspension. Nevertheless, because of the parallel orientation of two of the gimbals' axes there is no gimbal available to accommodate rotation about one axis.

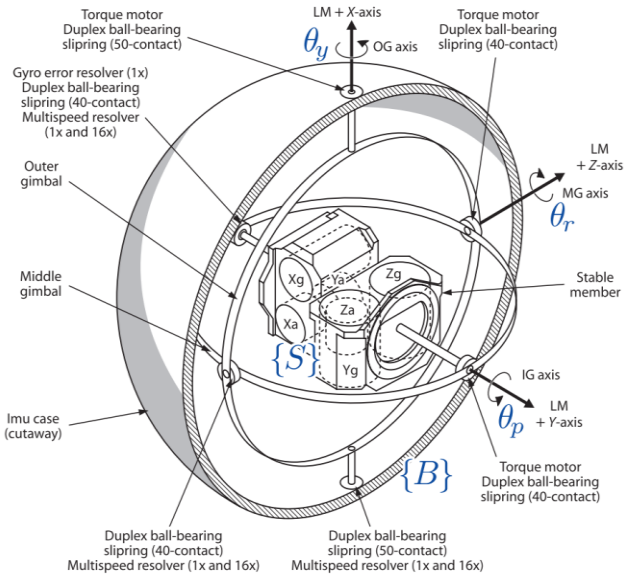
30

Gimbal Lock/Singularity



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NASA's Apollo IMU



Consider the situation when the rotation angle of the middle gimbal (rotation about the *z*-axis) is 90° – the axes of the inner and outer gimbals are aligned and they share the *same* rotation axis. Instead of the original three rotational axes, since two are parallel, there are now only two effective rotational axes – we say that one degree of freedom has been lost.



Image: NASA

A schematic of the inertial measurement unit, which uses a gimbal system to measure a spacecraft's velocity and attitude. NASA.

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Gimbal Lock

- The loss of a degree of freedom means that mathematically we cannot invert the transformation
- All three-angle representations of attitude, whether Eulerian or Cardanian, suffer this problem of
- The best that can be hoped for is that the singularity occurs for an attitude which does not occur during normal operation of the vehicle – it requires **judicious choice of angle sequence** and coordinate system.
- Singularities are an unfortunate consequence of using a minimal representation. To eliminate this problem, we need to adopt different representations of orientation.

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Credits

- This template is customized from a Slidesgo template.
- This lecture is made from the book “RVC by Professor Peter Corke”
- Some slides are inspired by Professor Peter Corke’s work.
- Rest of the material is self made and compiled from internet resources.

slidesgo

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